

AVNEET SINGH

Summary

Full-Stack Engineer & AI Enthusiast obsessed with the bridge between clean logic and machine intelligence. I specialize in **Python, React, and RAG architectures**, building scalable systems that work in the real world, not just on my machine. I enjoy solving complex engineering puzzles, optimizing **SQL workflows**, and ensuring my code is as efficient as my morning coffee routine.

Experience

The University of Toledo: College of Engineering

October 2023 - Present

IT System Administrator

Toledo, Ohio

- Managed and architected **Cloud-to-On-Premise data synchronization** for 500+ users, utilizing **Microsoft InTune** for endpoint management and **Active Directory (AD)** for granular identity and access management (IAM).
- Automated **high-volume log ingestion** and security auditing by developing **Python** and **PowerShell** scripts to monitor server integrity, reducing manual data-entry and ensuring 99.9% uptime for departmental **Cloud Servers**.

KPIT Technologies Limited

June 2025 - August 2025

Systems Engineer Intern

Novi, Michigan

- Developed modular communication interfaces using **C# and Python** to acquire real-time data from automotive hardware via serial and USB protocols.
- Architected a modular storage layer for real-time telemetry, utilizing **GitHub CI/CD** and branch protection to ensure the reliability and schema integrity of distributed data streams.
- Collaborated in an **Agile environment**, participating in daily scrums and code reviews to improve application reliability and performance.

KPIT Technologies Limited

June 2024 - August 2024

Software Engineer Intern

Novi, Michigan

- Engineered automated **ETL** tools using **Pandas** and **NumPy**, leveraging **vectorized transformations** to process large-scale diagnostic datasets and reducing manual reporting time by 70% .
- Developed **SQL**-driven data validation workflows to monitor ECU logic states, ensuring high data quality and reliability across complex middleware standards.

Softage Information Technology Limited

June 2023 - July 2023

Web Developer Intern

Gurugram, India

- Built a full-scale enterprise web platform using **Django, SQL** to streamline the training and fine-tuning of proprietary AI models for corporate clients, utilizing **RESTful API's** to improve data ingestion and model retraining by 20%.

Projects

RAG OpenPDF AI ChatBot | FastAPI, React, LangChain, OpenAI, FAISS

August 2025

- Engineered a full-stack AI application using **Python (FastAPI)** and **React**, enabling natural language querying of unstructured data through **Retrieval-Augmented Generation (RAG)**.
- Optimized document chunking and **vector similarity search** via **FAISS**, significantly improving retrieval accuracy.

Moodify – AI Mood-Based Music Player | React.js, FastAPI, PostgreSQL, Node.js, OpenCV

May 2025

- Architected a scalable data pipeline utilizing a **React frontend** to stream real-time facial feature metadata to a **FastAPI** backend hosted on **AWS EC2**, ensuring high availability and sub-second inference speeds.
- Designed and managed a relational PostgreSQL database to handle user session metadata and playlist state, utilizing AWS S3 for persistent storage of processed visual data and system logs.

Network Report Generation Tool | Python, Pandas, Numpy, PyQt5

August 2024

- Developed an automated **Python** analysis tool that utilized **SQL** for high-volume log retrieval and persistent storage, while managing the full development lifecycle via **GitHub** to ensure **version integrity** and technical specifications.

Education

The University of Toledo: College of Engineering

Anticipated Graduation Date: May 2026

Bachelor of Science: Computer Science and Engineering

Dean's List: Spring 2023, Spring 2024, Fall 2025

Skills

AI & Machine Learning: RAG, LangChain, OpenAI API, FAISS, Embedding Models, OpenCV, Computer Vision, YOLO
Data Engineering & Cloud: ETL/ELT Pipelines, AWS (S3, EC2, Lambda basics), Azure, Microsoft InTune, Active Directory, Data Modeling, Schema Design,
Languages: Python (Pandas, NumPy, Scapy), SQL (PostgreSQL, MySQL, Redshift), Java, C#, C, JavaScript, PowerShell
Frameworks & Libraries: FastAPI, Django, React, Node.js, Tailwind, Scapy, Tkinter
Tools & Cloud: Git, GitHub, Docker, Azure, VS Code, PyCharm, PowerShell, MySQL, MongoDB, SSMS
Soft Skills: Team Leadership, Collaborative Problem-solving, Adaptable, Technical Communication, Meticulous